

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study
GitHub Repository	https://github.com/sangee-3010/Machine_Learning
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1 Objective

- To implement a Decision Tree classifier.
- To extend the Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare single-tree and ensemble-tree models.

2 Problem Statement

Given 30 numerical features describing the nuclei of cells from a breast mass, the task is to build a binary classifier that predicts whether the mass is **Malignant (M)** or **Benign (B)**. Two supervised learning approaches – a single Decision Tree and a Random Forest ensemble – are trained, tuned via cross-validated grid search, and compared on a held-out test set using standard classification metrics.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC) Dataset
Dataset Source	UCI Machine Learning Repository (https://archive.ics.uci.edu)
Number of Samples	569
Number of Features	30 numerical attributes (mean, standard error, and worst value of 10 nuclear characteristics)
Number of Classes	2 (Malignant, Benign)
Missing Values	0
Train-Test Split	80% Train (455 samples) / 20% Test (114 samples), stratified, <code>random.state=42</code>

4 Exploratory Data Analysis

Class Distribution

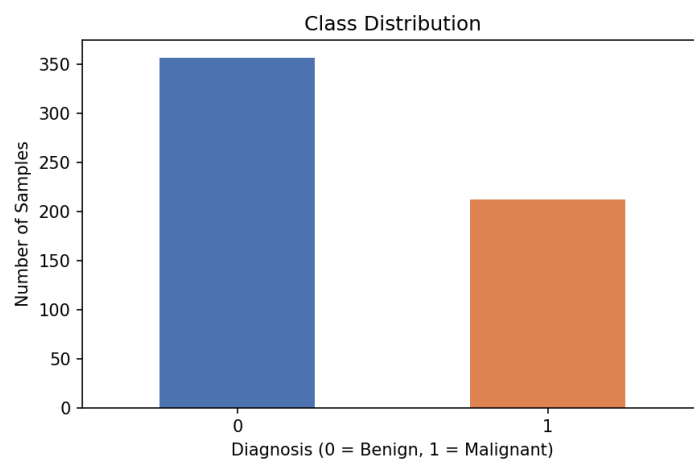


Figure 1: Class distribution of the target variable (0 = Benign, 1 = Malignant).

Inference: The dataset contains 357 Benign and 212 Malignant samples, giving a class ratio of roughly 63:37. While not perfectly balanced, the imbalance is mild, so stratified splitting and stratified cross-validation were used throughout to preserve class proportions in every fold and in the train/test partitions. This avoids a model that trivially favours the majority (Benign) class.

Feature Correlation

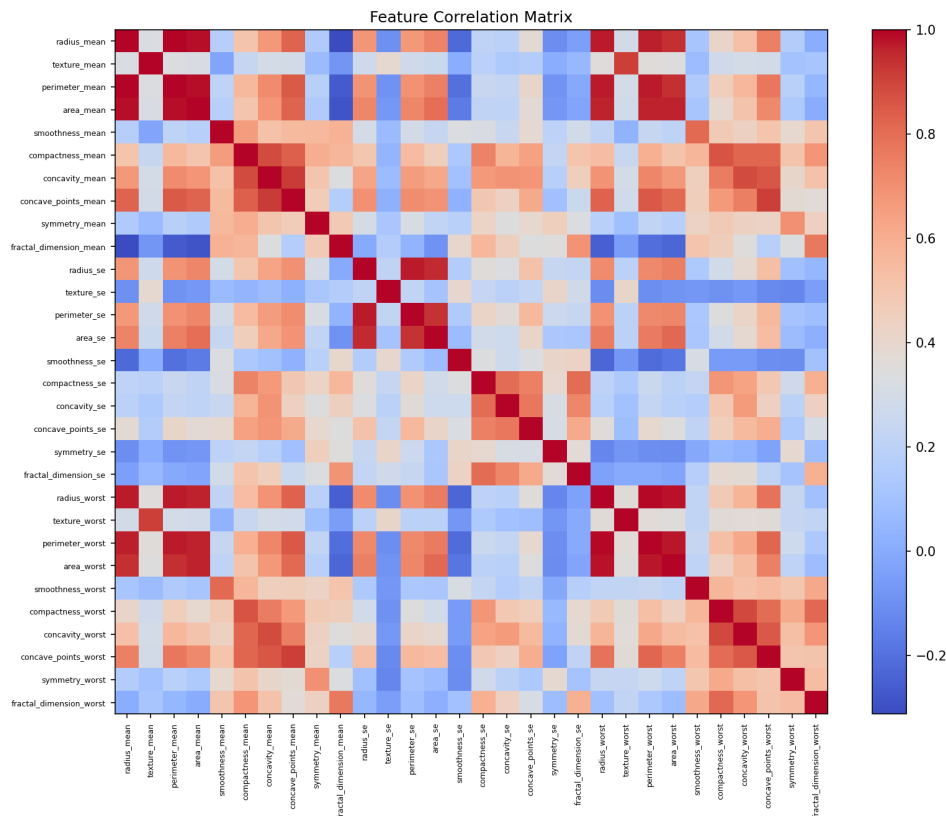


Figure 2: Pairwise Pearson correlation matrix of the 30 numerical features.

Inference: Strong positive correlation is visible among the size-related features (`radius`, `perimeter`, `area` and their `_worst` counterparts), which is expected since they measure related aspects of tumour geometry. The `concave_points` and `concavity` features are also mutually correlated and, as later confirmed by the feature importance analysis, are strongly associated with malignancy. This redundancy suggests that tree-based models, which naturally perform implicit feature selection at each split, are well suited to this dataset, and that dimensionality reduction is not strictly necessary for either Decision Tree or Random Forest.

5 Data Preprocessing

- **Missing value handling:** None required; the dataset has zero missing values.
- **Encoding:** The categorical target `Diagnosis` was label-encoded, mapping Benign $\rightarrow 0$ and Malignant $\rightarrow 1$.
- **Feature scaling:** Not applied. Both Decision Trees and Random Forests split on feature thresholds rather than distances, so they are scale-invariant and do not require normalization or standardization.
- **Train-test split:** An 80–20 stratified split (`random_state=42`) was used to preserve the Benign/Malignant ratio in both partitions.

- **Feature engineering:** Not applied; all 30 original numerical features were retained.

6 Machine Learning Algorithm

Decision Tree Classifier

A Decision Tree recursively partitions the feature space by selecting, at each node, the feature and threshold that most reduce class impurity (measured by Gini index or entropy). Splitting continues until a stopping criterion (maximum depth, minimum samples per leaf/split) is met, and predictions are made by traversing the tree to a leaf.

$$\text{Entropy}(S) = - \sum_{i=1}^c p_i \log_2 p_i, \quad \text{Gini}(S) = 1 - \sum_{i=1}^c p_i^2$$

Advantages: Highly interpretable, requires no feature scaling, handles non-linear decision boundaries naturally.

Limitations: Prone to high variance and overfitting, especially as depth increases; small changes in the training data can produce a very different tree.

Random Forest Classifier

A Random Forest is an ensemble of Decision Trees, each trained on a bootstrap-resampled subset of the training data (bagging), with each split further restricted to a random subset of features. Final predictions are obtained by majority vote across all trees. **Advantages:** Averaging over many decorrelated trees reduces variance and improves generalization compared to a single tree, while retaining feature importance interpretability.

Limitations: Less interpretable than a single tree, more computationally expensive to train and tune, and can still overfit if trees are too deep and too numerous relative to the data.

7 Experimental Setup

Python Version	3.x
Scikit-Learn Version	1.x
IDE	Jupyter Notebook
Operating System	—
Hardware	—

8 Hyperparameter Selection

Both models were tuned using `GridSearchCV` with 5-fold `StratifiedKFold` cross-validation (`random_state=42`), optimizing the F1-score.

Decision Tree Search Space and Best Parameters

Parameter	Value
Search space: criterion	{gini, entropy}
Search space: max_depth	{2, 3, 4, 5, 6, 8, 10, None}
Search space: min_samples_split	{2, 5, 10}
Search space: min_samples_leaf	{1, 2, 4}
Best: criterion	entropy
Best: max_depth	4
Best: min_samples_split	2
Best: min_samples_leaf	2
Best CV F1-score	0.9163

Random Forest Search Space and Best Parameters

Parameter	Value
Search space: n_estimators	{50, 100, 150}
Search space: max_depth	{3, 5, 8, 10, None}
Search space: max_features	{sqrt, log2}
Search space: bootstrap	{True, False}
Best: n_estimators	50
Best: max_depth	10
Best: max_features	sqrt
Best: bootstrap	False
Best CV F1-score	0.9617

Top Hyperparameter Combinations

Criterion	Max Depth	Min Split	Min Leaf	Avg CV Accuracy (%)	Avg CV F1 (%)
entropy	4	2	2	94.07	91.63
entropy	4	5	2	94.07	91.63
entropy	4	2	1	93.85	91.45
entropy	4	10	2	93.85	91.42
entropy	5	10	2	93.63	91.21

Table 2: Table 1: Decision Tree Hyperparameter Evaluation (Top 5 of 5-Fold CV grid search)

n_estimators	Max Depth	Max Features	Bootstrap	Avg CV Accuracy (%)	Avg CV F1 (%)
50	10	sqrt	False	97.14	96.17
150	8	log2	False	97.14	96.15
50	None	sqrt	False	96.92	95.89
150	10	log2	False	96.92	95.87

150	None	log2	False	96.92	95.87
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Table 3: Table 2: Random Forest Hyperparameter Evaluation (Top 5 of 5-Fold CV grid search)

9 Results

Decision Tree

Metric	Value
Accuracy	0.9298
Precision	1.0000
Recall	0.8095
F1-score	0.8947
ROC-AUC	0.9563

Random Forest

Metric	Value
Accuracy	0.9649
Precision	1.0000
Recall	0.9048
F1-score	0.9500
ROC-AUC	0.9950

5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	93.41	95.60	90.11	95.60	95.60	94.07
Random Forest	96.70	98.90	96.70	95.60	97.80	97.14

Table 4: Table 3: 5-Fold Cross-Validation Accuracy Comparison (%)

10 Result Visualization

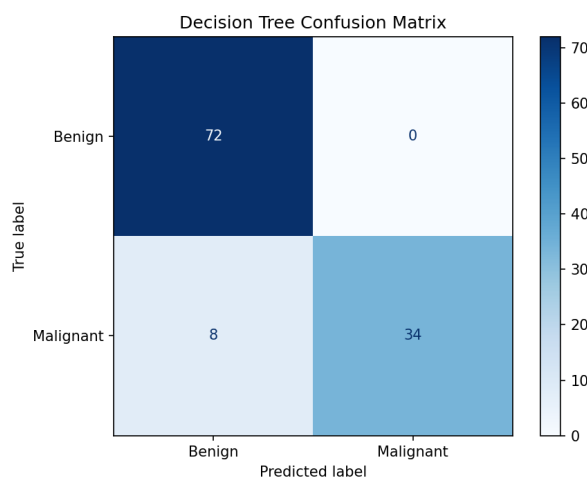


Figure 3: Decision Tree confusion matrix on the test set.

Inference: The Decision Tree correctly classifies all 72 Benign cases (no false positives, precision = 1.00) but misses 8 of the 42 Malignant cases, classifying them as Benign. In a diagnostic setting these false negatives are the more costly error type, since a missed malignancy delays treatment far more than a false alarm.

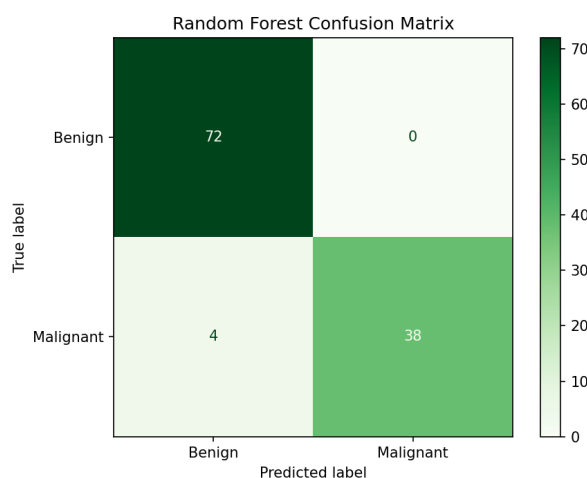


Figure 4: Random Forest confusion matrix on the test set.

Inference: The Random Forest also achieves perfect precision (no Benign case is misclassified as Malignant) while reducing false negatives from 8 to 4 compared to the Decision Tree, correctly identifying 38 of the 42 Malignant cases. This directly reflects the recall improvement from 0.8095 to 0.9048.

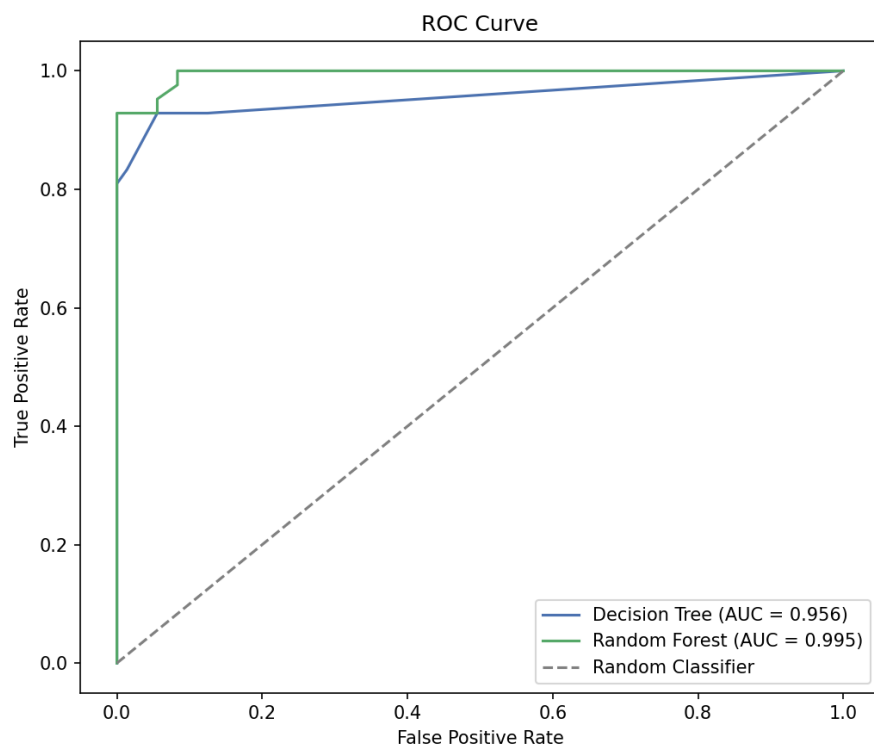


Figure 5: ROC curves for the Decision Tree and Random Forest classifiers.

Inference: The Random Forest ROC curve dominates the Decision Tree curve almost everywhere, achieving an AUC of 0.9950 versus 0.9563. This indicates the ensemble separates the two classes much more reliably across all classification thresholds, not just at the default 0.5 cut-off used for the confusion matrices above.

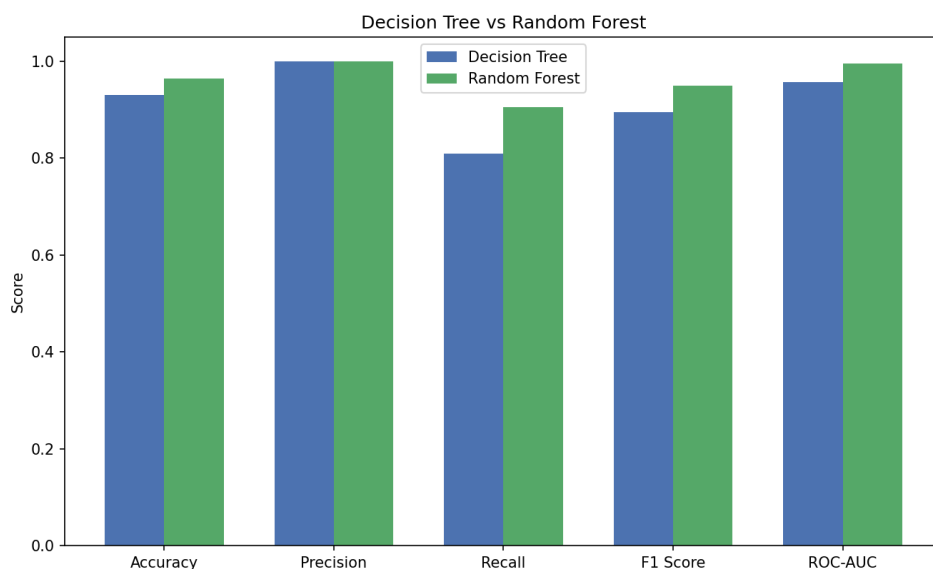


Figure 6: Side-by-side comparison of Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Inference: Random Forest matches the Decision Tree on precision but outperforms it on every other metric, most notably recall and ROC-AUC. The gap is largest for ROC-

AUC, confirming that ensembling improves the model’s ranking/probability quality, not just its hard-label accuracy.

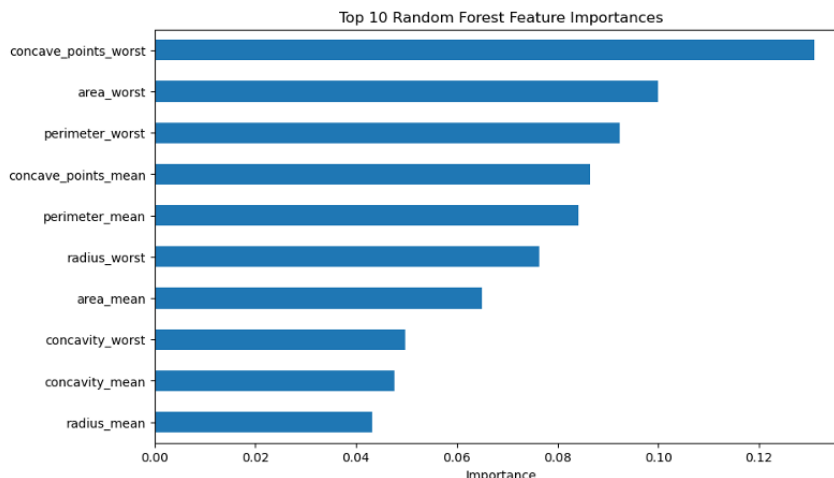


Figure 7: Top 10 most important features for the Random Forest model.

Inference: `concave_points_worst`, `area_worst`, and `perimeter_worst` are the most influential predictors, consistent with the Decision Tree’s own top split on `perimeter_worst`. Unlike the Decision Tree, whose importances are concentrated almost entirely on a handful of features (`perimeter_worst` alone accounts for over 66%), the Random Forest distributes importance more evenly across a broader set of correlated size- and shape-related features, which is a direct consequence of its random feature subsampling at each split.

11 Observations

1. **Effect of tree depth on overfitting:** Increasing the depth of a Decision Tree increases its complexity and allows it to fit the training data more closely. Very deep trees can capture noise and therefore carry a higher risk of overfitting. In this experiment, the tuned Decision Tree selected a maximum depth of 4, providing a good balance between model complexity and generalization – deeper trees in the grid search did not yield a better cross-validated F1-score.
2. **Hyperparameter with the greatest impact:** For the Decision Tree, `max_depth` had the most noticeable influence on cross-validated performance, since it directly controls model complexity; `criterion`, `min_samples_split`, and `min_samples_leaf` produced only small secondary adjustments. For the Random Forest, `max_depth` and `bootstrap` had the largest effect, with `max_features` and `n_estimators` playing a smaller supporting role.
3. **How Random Forest improves generalization:** Random Forest improves generalization by aggregating predictions from many trees trained on different resamples of the data and different random subsets of features at each split. This decorrelates the individual trees, so errors specific to any one tree tend to average out, producing lower-variance, more stable predictions than a single Decision Tree.

4. **Did ensemble learning always improve performance?** In this experiment, yes – the Random Forest improved every metric over the Decision Tree except precision, where both models tied at a perfect 1.00. Test accuracy increased from 92.98% to 96.49%, F1-score from 0.8947 to 0.9500, and ROC-AUC from 0.9563 to 0.9950. This is not guaranteed in general: ensembling helps most when the base learner has high variance, as a single decision tree does; if the base model were already low-variance and well-regularized, or if the dataset were very small or noisy, bagging would yield a smaller or negligible improvement.

12 Discussion

Both models achieve strong performance on this dataset, but the Random Forest ensemble consistently outperforms the single Decision Tree across every evaluation metric except precision, where they are tied at a perfect 1.00. The single Decision Tree, even after hyperparameter tuning (`max_depth=4`), shows higher variance across cross-validation folds (ranging from 90.11% to 95.60% accuracy) compared to the Random Forest (95.60% to 98.90%), illustrating the bias-variance trade-off: a shallow tree with only a handful of splits is more sensitive to which particular folds it sees, whereas averaging over 50 trees smooths out this fold-to-fold variability. The main limitation of the Decision Tree here is recall on the Malignant class (80.95%), which the Random Forest improves to 90.48% by combining evidence from many decorrelated trees rather than relying on a single greedy sequence of splits. Note that while Random Forest is conventionally described as using bootstrap aggregation, the grid search in this experiment allowed `bootstrap` to be either `True` or `False`, and the best-performing configuration selected `bootstrap=False` (each tree trained on the full training set rather than a bootstrap resample); the variance reduction here therefore comes primarily from random feature subsampling (`max_features=sqrt`) at each split rather than from bagging itself. A possible improvement for both models would be to explicitly optimize the decision threshold (rather than the default 0.5) to further reduce false negatives, given the asymmetric cost of missing a malignant diagnosis.

13 Conclusion

A Decision Tree classifier and a Random Forest ensemble were implemented on the Wisconsin Diagnostic Breast Cancer dataset and tuned using 5-fold cross-validated grid search over their respective hyperparameter spaces. The tuned Decision Tree (`criterion=entropy`, `max_depth=4`) achieved 92.98% test accuracy and an F1-score of 0.8947, while the tuned Random Forest (`n_estimators=50`, `max_depth=10`, `max_features=sqrt`, `bootstrap=False`) achieved 96.49% test accuracy and an F1-score of 0.9500, with a markedly higher ROC-AUC (0.9950 vs. 0.9563). These results confirm that ensembling via bagging and random feature selection reduces variance and improves generalization relative to a single decision tree, and that in this experiment ensemble learning did improve performance on every metric evaluated.

14 References

1. Scikit-learn Developers, “Decision Trees,” *scikit-learn documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/tree.html>
2. Scikit-learn Developers, “Ensemble methods: Random Forests,” *scikit-learn documentation*. [Online]. Available: <https://scikit-learn.org/stable/modules/ensemble.html>
3. W. Wolberg, O. Mangasarian, N. Street, and W. Street, “Breast Cancer Wisconsin (Diagnostic) Data Set,” *UCI Machine Learning Repository*, 1995. [Online]. Available: <https://archive.ics.uci.edu>